

Physical Chemistry of Materials

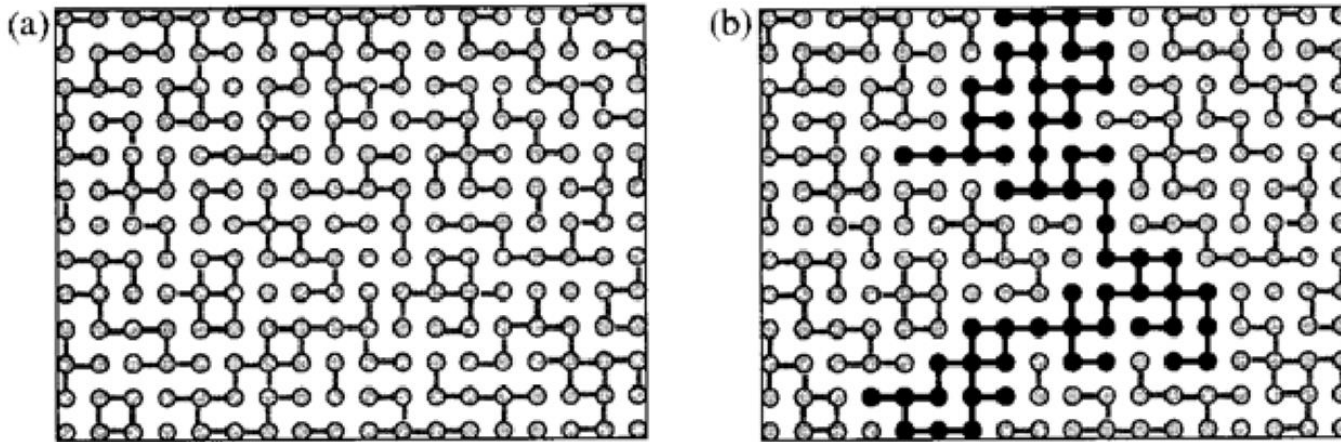
Lecture 22

Viscoelasticity of Polymers



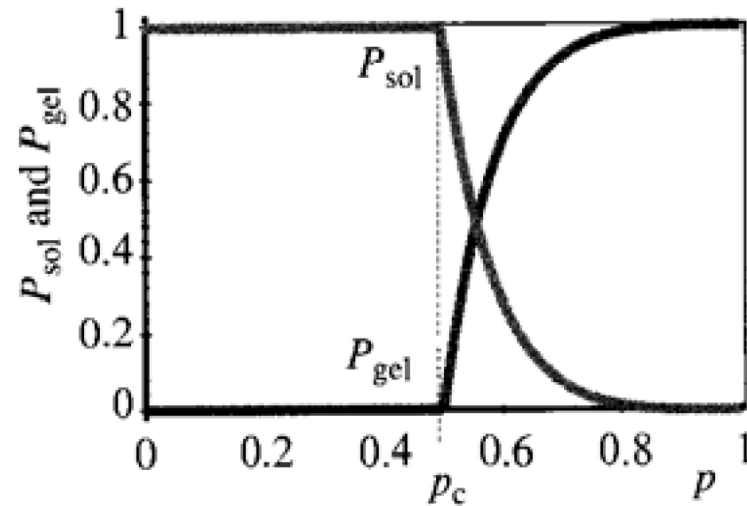
Last Class

Percolation: Network Formation



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 6.14)

Sol-Gel Transition



p = extent of reaction

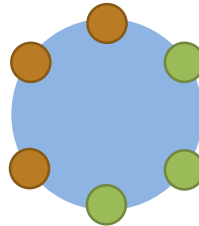
P_{sol} = “sol” fraction

P_{gel} = “gel” fraction

p_c = gel point

Flory-Stockmayer Theory

- Developed by Flory (1941) and Stockmayer (1944)
- A polymer with **functionality f** and **extent of reaction p**



- Construction of a network by probabilistic linkage
- Classical, mean-field theory of gelation

Flory-Stockmayer Theory

- Gel point

$$p_c = \frac{1}{f - 1}$$

- Sol fraction

$$P_{\text{sol}}^{1/f} = 1 - p + pP_{\text{sol}}^{(f-1)/f}$$

- Average number of polymers per cluster (for $p < p_c$)

$$N_n(p) = \frac{1}{1 - pf/2}$$

Critical Exponent

$$P_{\text{gel}} \sim (p - p_c)^\beta$$

- Flory-Stockmayer theory: $\beta = 1$ (works for $D \geq 6$)
- Computer simulation for 3D system: $\beta \approx 0.41$

FS theory only allows single-link inter-chain interactions

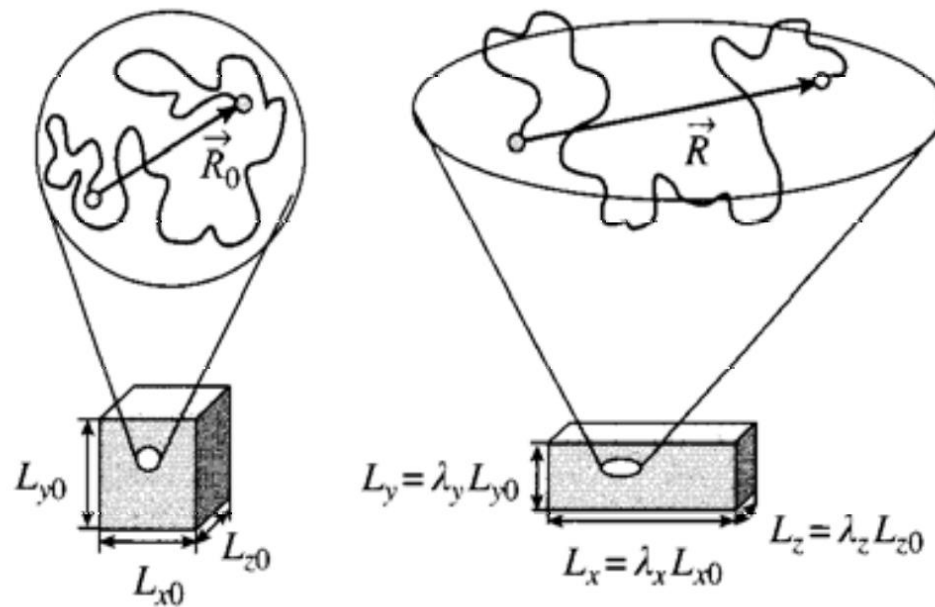
Rubber

- **Rubber:** polymer melt where crosslinks bond the chains together to form a macroscopic network



- “Liquid-like” state (temperature $> T_m$ or T_g)
- Bulk cannot flow due to crosslinks → “soft” solid

Affine Deformation



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 7.1)

Deformation Entropy

$$\vec{R}_0 = (x, y, z) \rightarrow \vec{R} = (\lambda_x x, \lambda_y y, \lambda_z z)$$

- End-to-end distances

$$R_0^2 = x^2 + y^2 + z^2$$

$$R^2 = \lambda_x^2 x^2 + \lambda_y^2 y^2 + \lambda_z^2 z^2$$

- Entropy $S = -3k_B R^2 / 2Na^2$ (cf. blob argument)

$$\Delta S = -\frac{3k_B}{2Na^2} [(\lambda_x^2 - 1)x^2 + (\lambda_y^2 - 1)y^2 + (\lambda_z^2 - 1)z^2]$$

Deformation Entropy

$$\Delta S = -\frac{3k_B}{2Na^2} [(\lambda_x^2 - 1)x^2 + (\lambda_y^2 - 1)y^2 + (\lambda_z^2 - 1)z^2]$$

If there are n polymers per unit volume, we can use the relation

$$\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle = Na^2/3$$

and obtain

$$\Delta S_{\text{volume}} = -\frac{nk_B}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

Deformation Helmholtz Energy

$$\Delta S_{\text{volume}} = -\frac{nk_{\text{B}}}{2}(\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

With no change in energy, Helmholtz energy change is

$$\Delta A_{\text{volume}} = \Delta U_{\text{volume}} - T\Delta S_{\text{volume}}$$

$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2}(\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

Uniaxial Elongation

$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

For incompressible rubber, $\lambda_x \lambda_y \lambda_z = 1$ (volume invariance)

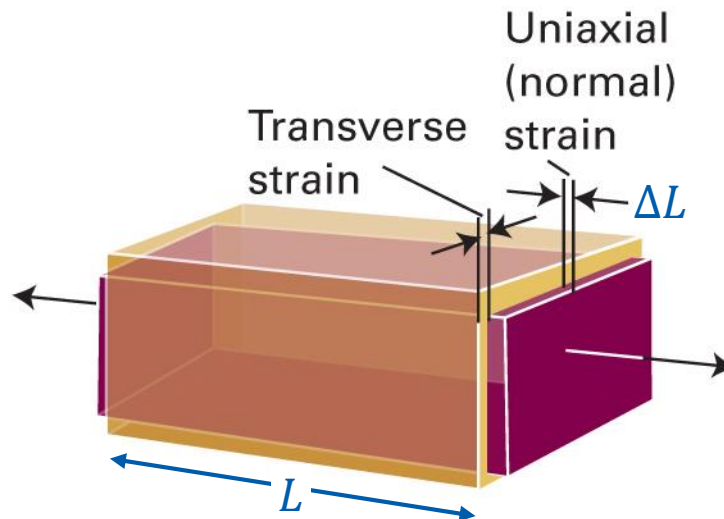
Simple elongation in the x direction by a factor of λ gives

$$\lambda_x = \lambda; \quad \lambda_y = \frac{1}{\sqrt{\lambda}}; \quad \lambda_z = \frac{1}{\sqrt{\lambda}}$$

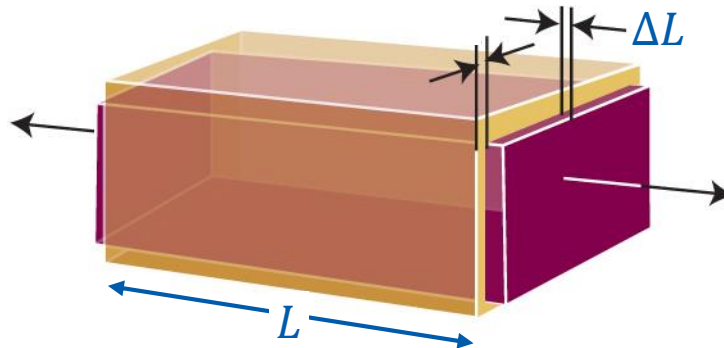
$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} \left(\lambda^2 + \frac{2}{\lambda} - 3 \right)$$

Young's Modulus

$$E = \frac{\text{uniaxial stress}}{\text{uniaxial strain}} = \frac{\tau}{e}$$



Uniaxial Strain



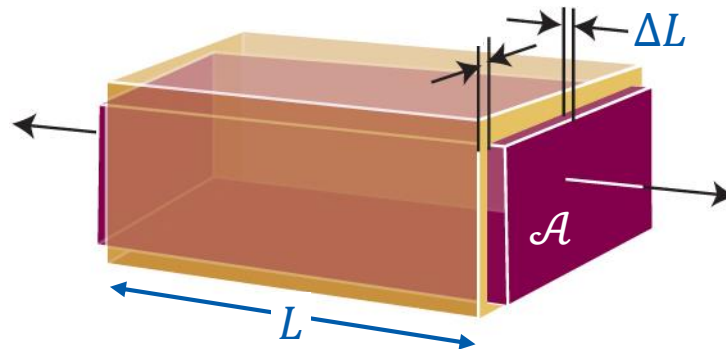
$$e = \frac{\Delta L}{L} = \frac{\lambda L - L}{L} = \lambda - 1$$

$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} \left(\lambda^2 + \frac{2}{\lambda} - 3 \right)$$

$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} \left[(1 + e)^2 + \frac{2}{1 + e} - 3 \right]$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 15D.3)

Uniaxial Stress



$$\tau = \frac{F}{A} = \frac{1}{A} \frac{d(V\Delta A_{\text{volume}})}{dx} = \frac{d\Delta A_{\text{volume}}}{d\lambda} = \frac{d\Delta A_{\text{volume}}}{de}$$

$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} \left[(1 + e)^2 + \frac{2}{1 + e} - 3 \right]$$

$$\tau = nk_{\text{B}}T \left[(1 + e) - \frac{1}{(1 + e)^2} \right]$$

Young's Modulus

$$E = \frac{\text{uniaxial stress}}{\text{uniaxial strain}} = \frac{\tau}{e}$$

$$\tau = nk_{\text{B}}T \left[(1 + e) - \frac{1}{(1 + e)^2} \right]$$

For small e ,

$$\tau \approx nk_{\text{B}}T[(1 + e) - (1 - 2e)] = 3nk_{\text{B}}Te$$

Thus,

$$E = 3nk_{\text{B}}T$$

Mechanical Properties of Rubber

For density ρ and average molecular mass M_x ,

- Young's modulus

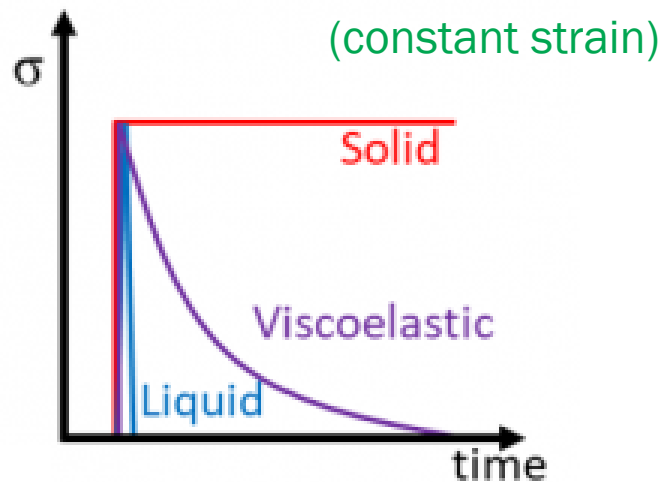
$$E = 3nk_{\text{B}}T = \frac{3\rho RT}{M_x}$$

- Shear modulus

$$G = nk_{\text{B}}T = \frac{\rho RT}{M_x}$$

Chemistry does not matter much!

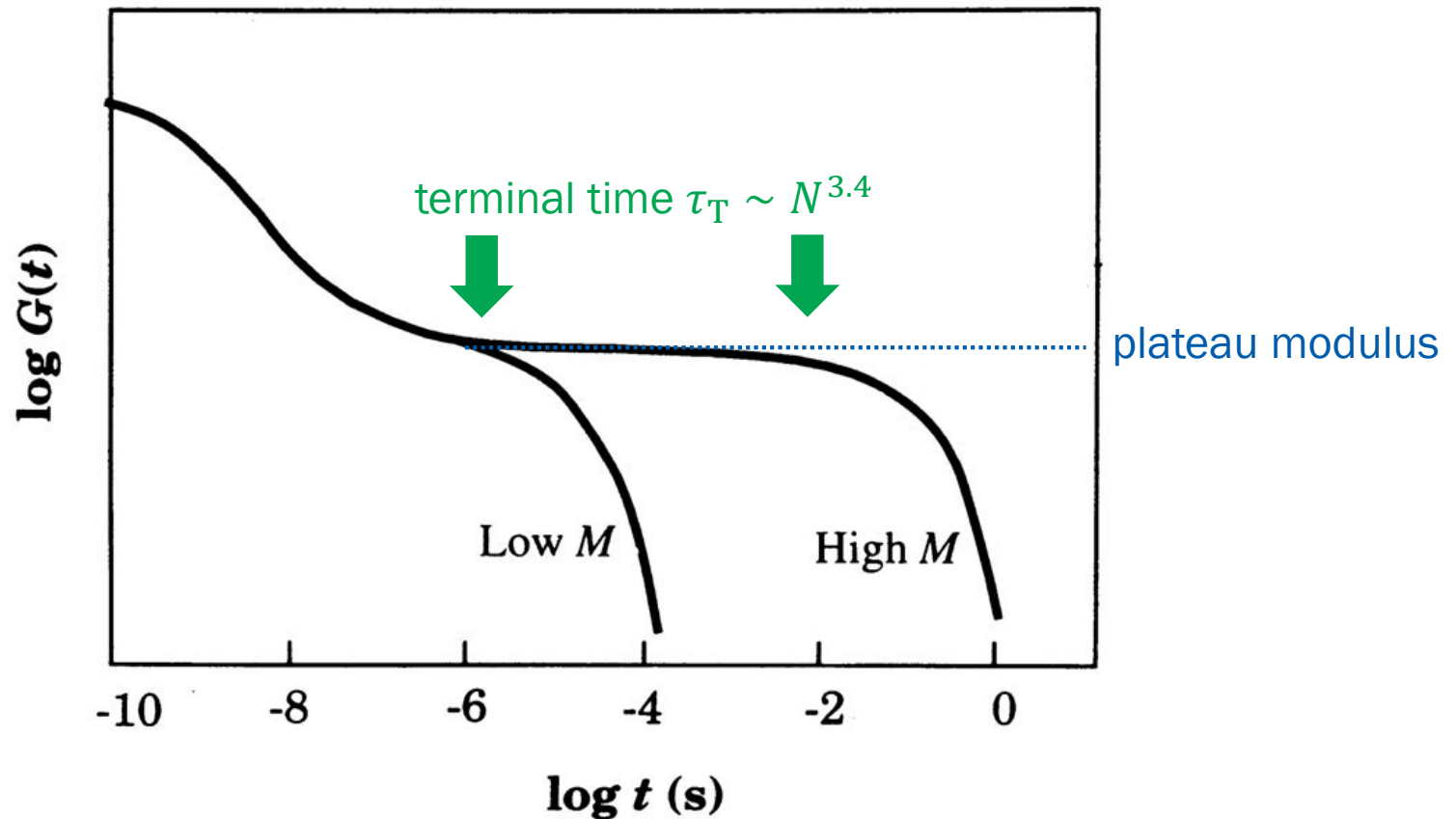
Viscoelasticity



- Stress relaxation modulus $G(t)$: constant strain e_0

$$G(t) = \frac{\sigma(t)}{e_0}$$

Viscoelasticity of Polymer Melts



(Image: https://link.springer.com/chapter/10.1007/978-3-030-27701-7_20)

Viscoelasticity of Polymer Melts

$t < \tau_T$: polymer behaves like an elastic solid

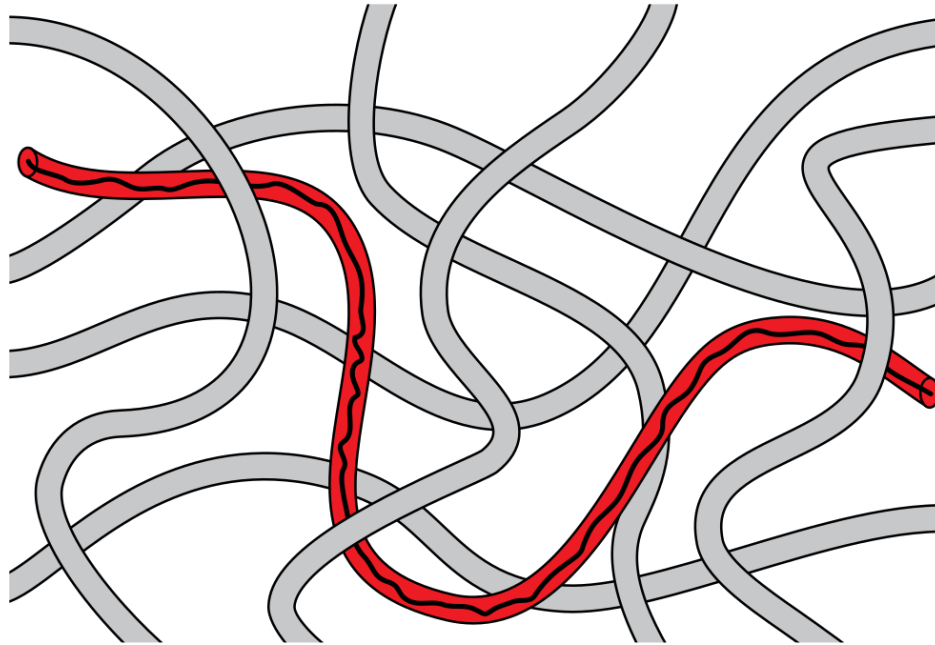
$t > \tau_T$: polymer behaves like a viscous fluid

Why do polymers behave like an elastic solid?

(Note: no crosslinks!)

Entanglement

- Affine deformation model: no topological constraint
- Real situation:



(Image: <https://commons.wikimedia.org/wiki/File:Reptation.svg>)

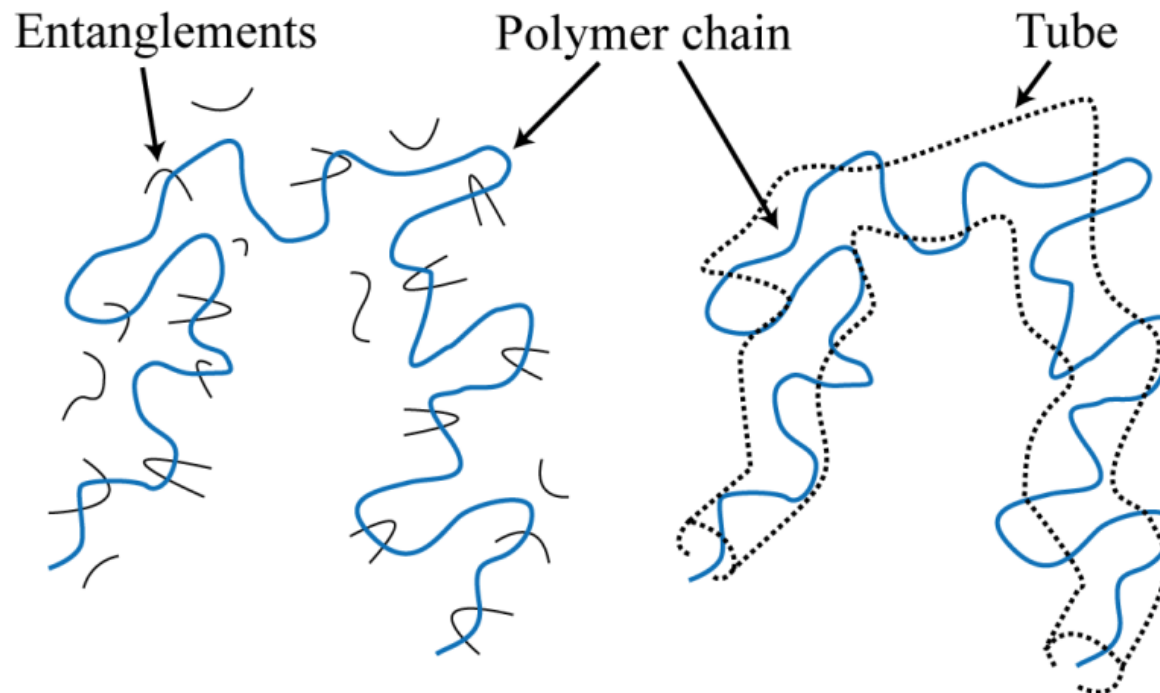
Origin of Elasticity

- Entanglements behave like temporary crosslinks
- Modulus (rubber model)

$$G \sim \frac{\rho RT}{M_e}$$

- ρ : density
- M_e : average molecular mass between entanglements
- Chains will disentangle with a characteristic timescale τ_T

Viscosity: Tube Model



slithering motion: “reptation”

Diffusion: Brownian Motion

- Motion of a particle in viscous medium

$$\vec{F} = \vec{F}_{\text{drag}} + \vec{F}_{\text{random}}$$

$$m \frac{d^2 \vec{r}}{dt^2} = -\xi \frac{d\vec{r}}{dt} + \vec{F}_{\text{random}}$$

where ξ is the drag coefficient.

For the x axis,

$$m \frac{d^2 x}{dt^2} = -\xi \frac{dx}{dt} + F_{\text{random}}$$

Rearrangement

$$m \frac{d^2 x}{dt^2} = -\xi \frac{dx}{dt} + F_{\text{random}}$$

$$\frac{\xi}{2} \frac{d(x^2)}{dt} = x F_{\text{random}} - m x \frac{d^2 x}{dt^2}$$

$$\frac{\xi}{2} \frac{d(x^2)}{dt} = x F_{\text{random}} - m \left[\frac{d}{dt} \left(x \frac{dx}{dt} \right) - \left(\frac{dx}{dt} \right)^2 \right]$$

$$\frac{\xi}{2} \frac{d\langle x^2 \rangle}{dt} = \langle x F_{\text{random}} \rangle - m \left[\frac{d}{dt} \left\langle x \frac{dx}{dt} \right\rangle - \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle \right]$$

Temperature Effect

$$\frac{\xi}{2} \frac{d\langle x^2 \rangle}{dt} = m \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = m \langle v^2 \rangle$$

From the equipartition theorem,

$$\frac{1}{2} m \langle v^2 \rangle = \frac{1}{2} k_B T$$

Hence,

$$\frac{d\langle x^2 \rangle}{dt} = \frac{2k_B T}{\xi}$$

Einstein Relation

$$\frac{d\langle x^2 \rangle}{dt} = \frac{2k_B T}{\xi}$$

Now, as $\langle r^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle + \langle z^2 \rangle = 3\langle x^2 \rangle$,

$$\frac{d\langle r^2 \rangle}{dt} = \frac{6k_B T}{\xi}$$

$$\langle r^2 \rangle = \frac{6k_B T}{\xi} t = \underline{D} t$$

diffusion coefficient

$$D = \frac{6k_B T}{\xi}$$

Diffusion by Reptation

- Diffusion coefficient of a chain

$$D = \frac{6}{\xi_{\text{tube}}} k_B T = \frac{6}{g_{\text{seg}} \xi_{\text{seg}}} k_B T$$

where ξ_{seg} is the drag coefficient inside a segment and g_{seg} is the number of segments.

As $g_{\text{seg}} \propto N$,

$$D \propto \frac{k_B T}{N}$$

Terminal Time

$$D \propto \frac{k_B T}{N}$$

As $\langle r^2 \rangle = Dt$, for a tube with length of $L \propto N$,

$$L^2 = D\tau_T$$

Hence,

$$\tau_T = \frac{L^2}{D} \propto N^{-3}$$

(Experimentally $\tau_T \sim N^{-3.4}$)

Self-Diffusion Coefficient

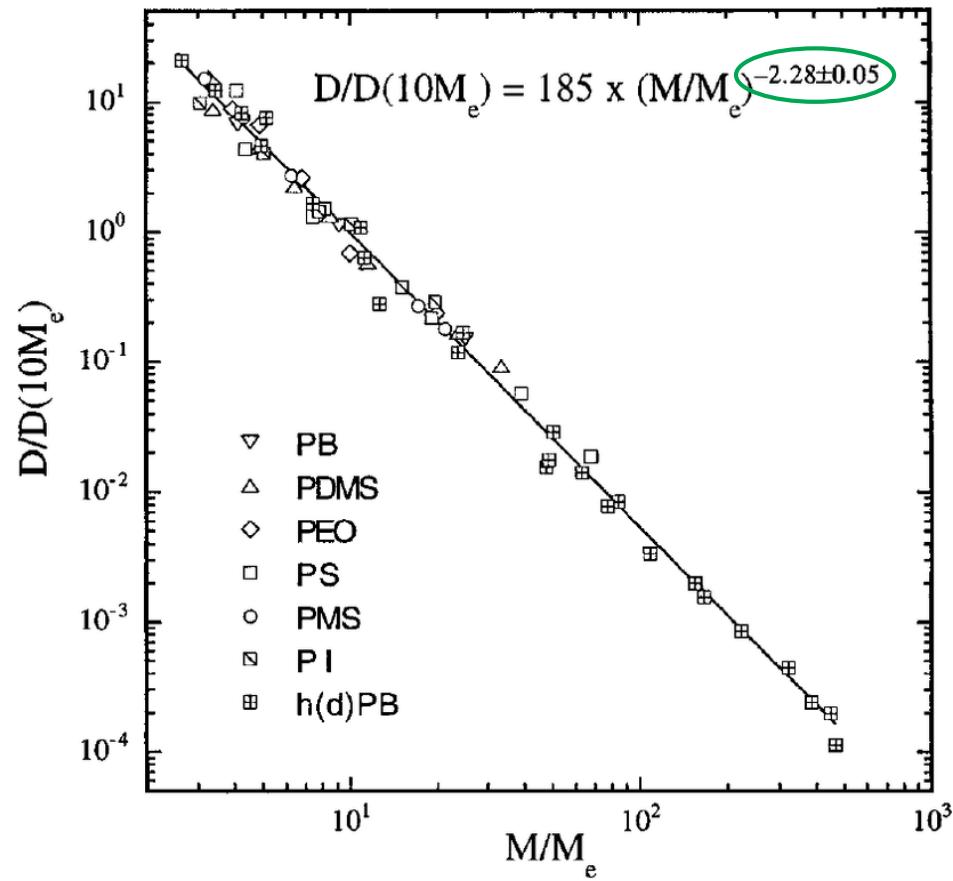
- Each chain diffuses a distance comparable to its own end-to-end distance during the terminal time
- End-to-end distance of a polymer in polymer melt

$$R \sim \sqrt{N}a \text{ (“ideal chain”)}$$

- Self-diffusion coefficient

$$D_{\text{self}} \sim \frac{Na^2}{\tau_T} \sim N^{-2}$$

Experimental Data



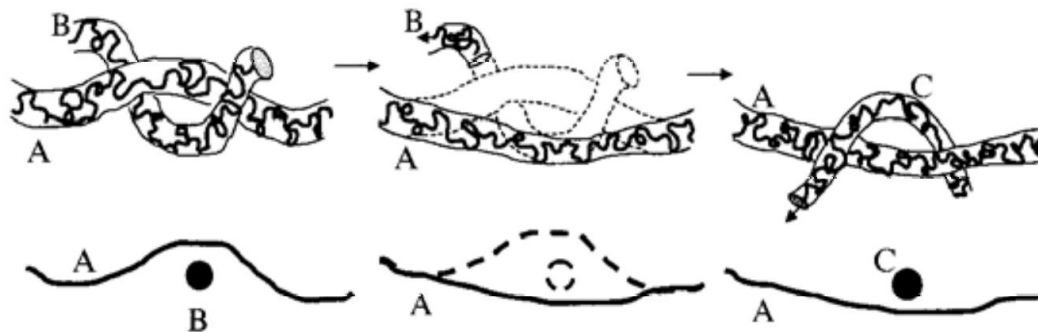
(Image: Karatrantos et al., *Polymers* (2019))

Reasons for Discrepancy

- Fluctuation of end-to-end distance



- Reptation of neighboring chains



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Figs. 9.1 and 9.25)